

FPGA HW – Input/Output Subsystem

The homework task is to implement the echo application using an interrupt driven approach.

Objectives

Upon successful completion of the homework task you should be able to:

1. Describe how an interrupt-driven input/output subsystem is configured and operated.
2. Validate in hardware that I/O is interrupt driven and the main loop is freed for useful work.

Implementation guidelines

Go back to Vivado, and connect the interrupt line of the debug module to the processor. Generate the bitstream and update the hardware platform in Vitis with the new .xsa file.

Create a new application project, and afterwards copy the accompanying `main.c` file from the `src` directory of the `interrupt_io` application project. The interrupt service routing (ISR) is already defined for you.

1. Using the the UART Product Guide [2] configure the peripheral to enable interrupt generation.
2. Using the Microblaze Reference Guide [1] configure MicroBlaze to respond to interrupts. Hint: A new header file has been included, what is its purpose?
3. Complete the ISR and relieve the while loop from any I/O tasks.

Use the debugger to confirm that the application is indeed interrupt-driven.

Deliverables

FPGA hardware demonstration showing both polled and interrupt-driven IO configurations.

One page report containing:

1. A description of the conceptual difference between polled and interrupt io.
2. A description of your implementation methodology. Visuals like flow charts are welcome.
3. The code included as an appendix.

References and appendices are excluded from the report length.

Rubric

The marks for this task are allocated as follows:

1. 50% for demonstration split equally between the polling lab task and interrupt homework.
2. 20% is allocated for the accuracy of technical content in the report

Lab #3.1 – FPGA, Input/Output Subsystem – Interrupt-driven IO

3. 20% is allocated for the code
4. 10% is good presentation and report writing.

This part of the homework contributes 10% towards the final course grade.

References

1. MicroBlaze Processor Reference Guide ([UG984](#))
2. MicroBlaze Debug Module LogiCORE IP Product Guide ([PG115](#))